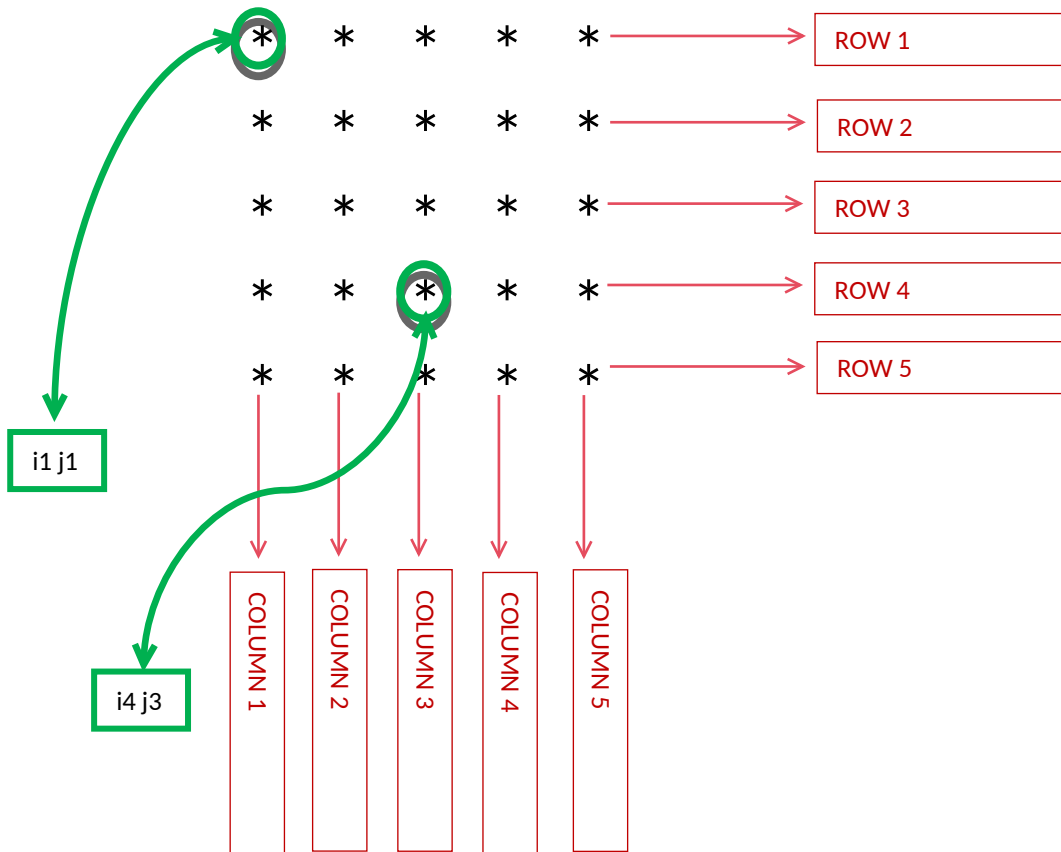


WEEK 01 MODULE
Programming Practice

TOPIC : 1 'Star Pattern'

1. SQUARE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a perfect square.
- number of rows is equal to the number of columns
- Every position within the square contains the same character (*).
- There are no empty spaces, boundary checks, increasing sequences, decreasing sequences, or special conditions involved.

ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row generated by the outer loop, traverse all columns of that row using the inner loop.

-The inner loop is responsible for generating columns.

Step 4: At each column position, print a star (*) because every cell of the square must be filled.

Step 5: After all columns of the current row have been processed, move the cursor to the next line.

Step 6: Repeat the same process until all n rows have been completed.

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class StarSquare {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the square: ");
        int n = sc.nextInt();

        for(int i = 0; i <= n-1; i++) {

            for(int j = 0; j <= n-1; j++) {
                System.out.print("* ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

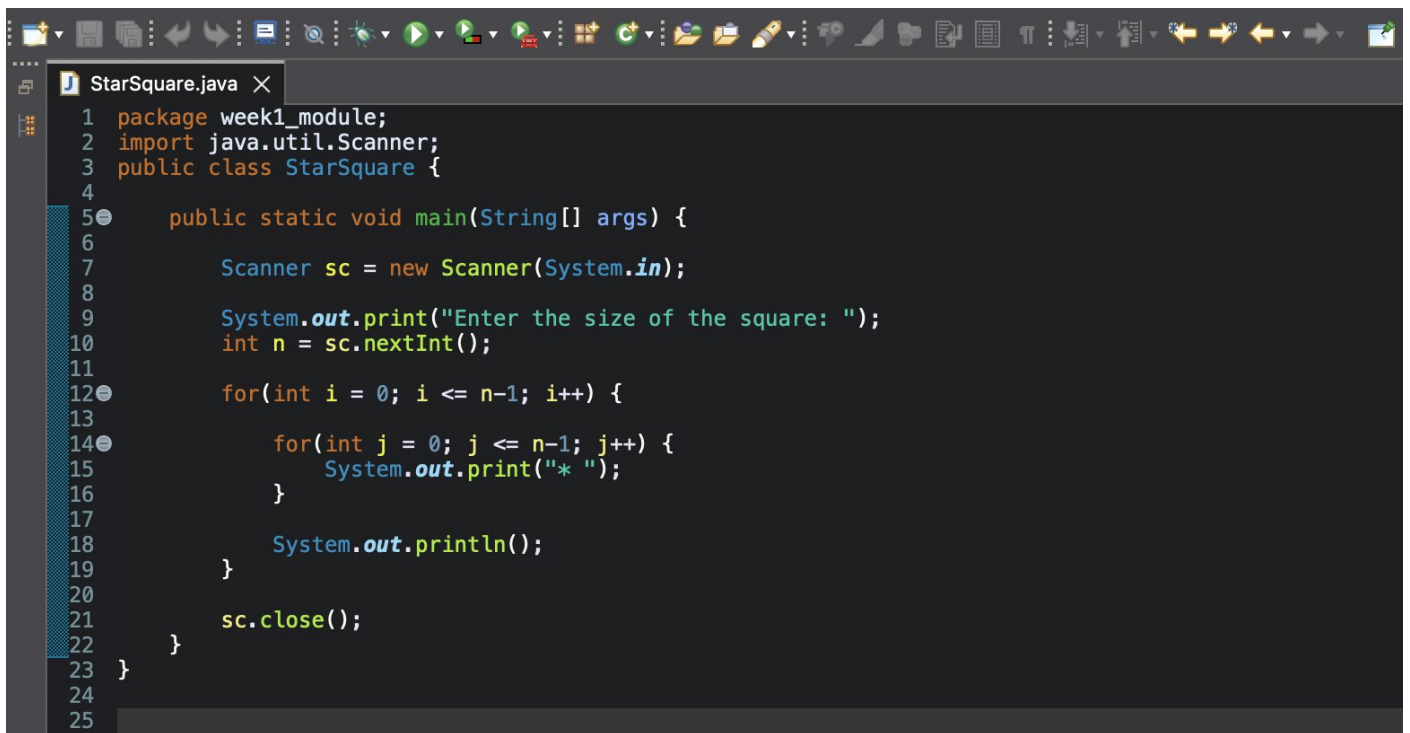
3. Program Output :-

Enter the size of the square: 5

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class StarSquare {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the size of the square: ");
10        int n = sc.nextInt();
11
12        for(int i = 0; i <= n-1; i++) {
13
14            for(int j = 0; j <= n-1; j++) {
15                System.out.print("* ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
25
```

ii. Program's Output :-



```
<terminated> StarSquare [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java (12-Jun-2026, 7:53:42 pm - 7:53:45 pm elapsed 0:00:03) [pid: 3633]
Enter the size of the square: 5
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```